

Vodacom M-Pesa Tanzania – Open API

Callbacks & Webhooks Documentation

Asynchronous Result Handling • Version 1.0 • September 2026

1. Overview

The M-Pesa Open API supports both **synchronous** and **asynchronous** modes. In synchronous mode you receive an immediate response (usually **INS-0** meaning the request was accepted). The final outcome (customer entered PIN, cancelled, or timed out) is delivered later via a **callback (webhook)** to the Response URL you configure in the portal.

Typical flow:

1. Your system calls C2B / B2C / B2B / Reversal
2. You receive immediate synchronous response (INS-0 = accepted)
3. Customer interacts with the USSD Push on their phone
4. M-Pesa POSTs the final result to your registered Response URL

2. Configuring the Response URL

1. Log in to <https://openapiportal.m-pesa.com>
2. Open your Application
3. Enable the required product (e.g. C2B Single Payment)
4. Locate the field **Response URL** or **Asynchronous Response URL**
5. Enter a publicly reachable **HTTPS** endpoint, for example:
`https://yourdomain.com/api/mpesa/callback`
6. Save the application

Requirements: HTTPS only • Publicly accessible (no localhost) • Stable uptime recommended. During development use ngrok, Cloudflare Tunnel, or similar tools.

3. Callback Payload Structure

When a transaction reaches a final state, M-Pesa sends an HTTP **POST** request with a JSON body to your Response URL.

Example Callback Body

```
{
  "input_OriginalConversationID": "1e029550d09745e7b2221bb4b2dc8ffc",
  "input_ThirdPartyReference": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "input_TransactionID": "RvvsqB0rcP3Y",
  "input_ResultCode": "0",
  "input_ResultDesc": "Request Processed Successfully"
}
```

Field Descriptions

Field	Description
input_OriginalConversationID	Conversation ID returned in the original synchronous response
input_ThirdPartyReference	The ThirdPartyConversationID that YOU sent in the original request. Use this to match the callback to your internal record
input_TransactionID	Final M-Pesa transaction identifier
input_ResultCode	"0" = Success. Any other value indicates failure, cancellation or timeout
input_ResultDesc	Human-readable description of the result

4. Required Response from Your Endpoint

Your server must respond quickly (ideally within a few seconds). Recommended acknowledgement:

```
{
  "output_OriginalConversationID": "1e029550d09745e7b2221bb4b2dc8ffc",
  "output_ResponseCode": "0",
  "output_ResponseDesc": "Successfully Accepted Result",
  "output_ThirdPartyConversationID": "a1b2c3d4-e5f6-7890-abcd-ef1234567890"
}
```

```
}
```

Returning HTTP **200 OK** is also accepted by most implementations. If your endpoint returns an error or times out, M-Pesa may retry the callback.

5. Recommended Handling Pattern

1. **Receive & log** the raw payload immediately (headers + body)
2. **Respond with 200** as fast as possible
3. Look up your internal transaction using **input_ThirdPartyReference**
4. Perform an **idempotency check** (have you already processed this TransactionID?)
5. Update your database status (SUCCESS / FAILED / CANCELLED)
6. Trigger business logic (fulfill order, send SMS/email, etc.) asynchronously if needed

6. Sample Implementation (Node.js / Express)

```
app.post('/api/mpesa/callback', async (req, res) => {
  const payload = req.body;

  // 1. Acknowledge immediately
  res.status(200).json({
    output_ResponseCode: "0",
    output_ResponseDesc: "Successfully Accepted Result",
    output_OriginalConversationID: payload.input_OriginalConversationID,
    output_ThirdPartyConversationID: payload.input_ThirdPartyReference
  });

  // 2. Process asynchronously
  try {
    const ref = payload.input_ThirdPartyReference;
    const txId = payload.input_TransactionID;
    const resultCode = payload.input_ResultCode;

    const payment = await Payment.findOne({ thirdPartyId: ref });
    if (!payment) return; // unknown reference

    // Idempotency
    if (['SUCCESS', 'FAILED'].includes(payment.status)) return;

    if (resultCode === "0") {
      payment.status = 'SUCCESS';
      payment.mpesaTransactionId = txId;
      await payment.save();
      // fulfill order, notify customer...
    } else {
      payment.status = 'FAILED';
      payment.failureReason = payload.input_ResultDesc;
      await payment.save();
    }
  } catch (err) {
    console.error('Callback processing error:', err);
    // push to retry queue if needed
  }
});
```

7. Best Practices

Practice	Why it matters
Idempotency	M-Pesa may deliver the same callback more than once
Log everything	Raw body + headers are essential for debugging and audits
Respond fast	Do heavy work after sending the 200 response
Unique references	Always send a unique ThirdPartyConversationID so you can match callbacks reliably
Fallback polling	Also implement Query Transaction Status as a safety net for missed callbacks
HTTPS only	Portal rejects non-HTTPS URLs
Stable URL	Prefer a permanent domain; avoid free ngrok URLs in production

8. Testing Callbacks

Development

- Run your local server and expose it with ngrok: `ngrok http 3000`
- Copy the HTTPS URL into the portal Response URL field
- Trigger a C2B payment using sandbox MSISDN **000000000001**

Production

- Use a stable HTTPS domain with good uptime
- Monitor for missed callbacks and use the Query Transaction Status API as backup
- Keep detailed logs of every callback received

9. Interpreting input_ResultCode

input_ResultCode	Meaning	Recommended Action
"0"	Success – funds moved	Mark payment SUCCESS and fulfill order
Non-zero	Failed / Cancelled / Timeout	Mark FAILED, show retry option to customer

For more granular failure reasons, also inspect **input_ResultDesc** and cross-reference with the INS-xxx codes documented in the main API guide.

This document complements the full Open API documentation. Always verify the latest callback field names inside the Open API Portal after logging in.

Document generated on 2026-09-22 • Portal: <https://openapiportal.m-pesa.com>